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Best Practices in Data Governance, Preparation and Analytics (CA6003)

Assignment submission Guidelines

1.Introduction

Data is the foundation of any successful AI system. The preparation and governance of data constitute a fundamental component in the development of reliable AI and machine learning solutions. This process typically begins with the systematic profiling and analytical exploration of dataset characteristics to identify data quality issues, inconsistencies, anomalies, bias, and potentially misleading relationships. These findings should guide appropriate data cleaning, transformation, enrichment, and validation to ensure suitability for downstream learning tasks. To complete the data-to-insight workflow, a basic machine learning model is employed to evaluate and demonstrate how data preparation decisions influence model behaviour, performance, and robustness, and reliability.

Effective data governance, preparation, and analytics represent a convergence of multiple disciplines spanning data engineering, statistics, ethics, and applied machine learning. The process is inherently technical, requiring the ability to interpret, validate, and transform raw data using appropriate tools and methodologies. At the same time, it demands critical analytical judgment to distinguish genuine patterns from data quality problems, investigate non-obvious anomalies and relationships, detect bias and integrity issues, and avoid common analytical fallacies during exploratory analysis. Data preparation should therefore be treated as an evidence-driven process, where important interventions are supported by observations from the data and their consequences are critically evaluated.

Evidence of mastery lies in the ability to integrate these skills to justify data-centric decisions and communicate their impact clearly and rigorously. This group assignment (group size: 4; groups are pre-assigned) provides students with the opportunity to apply the principles and best practices covered in the course by systematically investigating raw data, designing robust data preparation workflows, and empirically demonstrating their importance within the AI lifecycle. The emphasis is on developing reliable and defensible data practices rather than maximising predictive accuracy.

2. Assignment Objective

This assignment is explicitly designed to assess the following learning outcomes. Upon successful completion, students will demonstrate the ability to:

1. Prepare raw data for machine learning by performing systematic data profiling, cleaning, validation, transformation, and enrichment using appropriate tools and techniques (e.g., handling missing values, encoding, feature engineering, and standardisation).

   (Students should justify major data preparation decisions based on evidence from the data and validate their impact where appropriate.)

2. Analyse datasets to detect and address bias, outliers, and data integrity issues, evaluate their impact on downstream model performance, and apply suitable mitigation strategies.

(Students should investigate the possible causes and significance of identified issues, apply suitable mitigation strategies, and evaluate their impact on downstream analysis and model behaviour.)

3. Design robust Exploratory Data Analysis (EDA) by interpreting summary statistics and visualisations, identifying analytical fallacies (e.g., correlation vs causation, Simpson's paradox), and applying strategies for accurate and meaningful data interpretation.
4. Empirically evaluate the impact of data preparation decisions using a basic, interpretable machine learning model. Students should compare minimally processed and systematically prepared datasets and demonstrate how individual preparation steps influence predictive performance, robustness, fairness, or interpretability. Improved accuracy is not necessarily expected; students should critically explain both improvements and deterioration in model behaviour.

   Note that the key purpose is to identify that ML is evidence, not the goal. The assignment should reinforce data best practices. Only basic, interpretable models allowed (Example: Linear / Logistic Regression, Decision Tree)

- Investigate and explain at least one non-obvious pattern, anomaly, inconsistency, bias, or potentially misleading relationship discovered through systematic data profiling and exploratory analysis.
- Students must compare minimally processed data vs systematically prepared data using the same basic modelling approach.
- Marks are awarded for quality of investigation, explanation, reasoning, justification, insight and NOT for accuracy chasing

The emphasis of this assignment is data-centric AI practice, ethical reasoning, and analytical clarity rather than model complexity or predictive accuracy.

3. Video Presentation

Presentation Coverage: Your group is required to produce a 12-minute (maximum) recorded video presentation, in which all group members must appear and present. The video should clearly communicate the investigative and analytical process, focusing on data governance, profiling, exploratory analysis, data preparation, and their impact on basic machine learning outcomes.

The presentation should address the assignment objectives, highlighting the key data quality issues and non-obvious findings identified, rationale for important data preparation decisions, comparison of minimally and systematically prepared data, and their impact on model behaviour. The presentation should conclude with the group's key contributions and reflection on what was novel, unexpected, or insightful. The emphasis should be on reasoning, evidence, and interpretation rather than code or model accuracy.

Presentation Format and Requirements

- The presentation must be delivered as a single continuous video of no more than 12 minutes. Content beyond 12 minutes will not be considered for evaluation.
- All group members must appear on screen and contribute to the presentation.

- Slides may be used to support the presentation but should be concise and visually clear. Excessively wordy slides will not be scored favourably.
- Detailed code and supporting analyses should be provided in the submitted Jupyter Notebook.

Originality and Use of Tools

All dataset analysis, data preparation, modelling, visualisation, interpretation, and presentation should demonstrate the group's understanding and intellectual ownership. Plagiarism or submission of non-original work without appropriate acknowledgement is considered a serious academic violation and may result in severe penalties.

Generative AI tools may be used as supporting tools (e.g., for code generation/debugging, analytical suggestions, or language assistance) and must be clearly acknowledged. Students are responsible for critically evaluating and verifying AI-generated outputs and must be able to explain and justify all code, analyses, data preparation decisions, and conclusions included in their submission.

The use of GenAI should be briefly declared in the Jupyter Notebook, stating the tool used, purpose of use, and how the output was verified.

Submission Instructions

On behalf of the group, the group leader (or any one group member) must submit:

1. A public YouTube link to the 12-minute presentation video
2. Jupyter Notebook (.ipynb format)
    1. Submit the notebook in. ipynb format only.
    2. The notebook must be well-structured and clearly written, including:
        1. Clear headings and sections
        2. Detailed comments and markdown explanations
        3. Properly explained data cleaning, preprocessing, and analysis steps
    3. Code should be readable, organized, and reproducible.

Submissions that are unclear, poorly documented, or incomplete may result in loss of marks.

Both items must be submitted via the NTULearn Blackboard Assignment portal by the stated deadline. Multiple submissions are permitted (up to three), but only the final submission before the deadline will be assessed. Students are strongly encouraged to submit only when the group is satisfied that the submission is final.

4. Peer Evaluation

Peer Evaluation Rubrics – At the end of the assignment, it is necessary for all students to assess the contributions of each team member based on:

1. **Teamwork**: demonstrating proactiveness in collaborating with team members and respect for each other.
2. **Quantity of work**: demonstrating fair share in the overall workload throughout the team project.

3. **Quality of work**: contributing ideas and research efforts that enhance the overall quality of the team's output.

Scoring – Peer evaluation exercise is **confidential** and will be carried out after the assignment submission deadline is over. All students must complete their peer evaluation before stipulated deadline. Failing which, they will receive **zero marks** for their peer evaluation component, regardless of what scores they received from their peers. Each student will give a rating scale between 1 to 5 on the above three assessment criteria for each of his or her group members. Please carefully note the rating rubric below. Do note that a score of "1" is not the best rating, but the worst.

Rating Scale	
5	Strongly agree (*best rating*)
4	Agree
3	Neutral
2	Disagree
1	Strongly disagree (*lowest rating*)

Managing Your Team – Setup regular meeting (face to face or online) to monitor assignment progress. If team members are not contributing to their allocated assigned task and responsibilities, do provide timely feedback to them. Encourage them to contribute and remind them that their individual marks in the course can be impacted by a poor peer evaluation rating from the teammates.

5. Deliverables and Deadlines

The table below outlines the various deliverables for Video Group Assignment and their respective weightage.

Deliverables	Submission Mode	Weightage	Deadline
Notebook and Video Presentation – Completed Video presentation (≤ 12 min) along with the Jupyter Notebook (.ipynb) documenting the work. **Note 1:** Upload the video to YouTube (or any related domain) as **Public** and include the group number when submitting the link. **Note 2:** Content beyond the 12-minute limit will not be considered. **Note 3:** The notebook must be well-structured, readable, and reproducible, with clear sections, comments/Markdown explanations, and properly documented data profiling, cleaning, preparation, and analysis.	Upload to NTULearn Assignment portal	90%	Before Sunday, (18 October@12 noon)
Peer Evaluation – Completed peer assessment for each of your teammates using the peer review system on NTULearn, which should appear in the **Assignment** folder.	NTULean peer review system	10%	Before Wednesday, (21 October@12 noon)

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Assessment Criteria for Video presentation

Presentation - Assessment Criteria	Weightage
Appropriateness –Appropriate selection and use of data analysis techniques and visualizations at each stage of the workflow, including dataset exploration, data profiling, cleaning, bias analysis, exploratory data analysis, and model evaluation.	25%
Correctness & Clarity - Technical correctness of work carried out across all stages of the project, including data preparation, transformation, bias mitigation, exploratory analysis, selection and application of basic machine learning models, and evaluation of results. Insights derived from the analysis should be clearly explained, logically reasoned, and accurately communicated. Assessment marks will be allocated according to the clarity with which each team member's contributions are distinguished and documented	25%
Data Interpretation and Analytics – Demonstration of sound analytical judgment in interpreting summary statistics, distributions, and relationships within the data. This includes the ability to identify redundancy, multicollinearity, skewness, and potential analytical fallacies, and to justify conclusions drawn from the analysis in a rigorous and evidence-based manner.	25%
Novelty and depth of data centric insights – Originality and depth in the choice of problem context, dataset, data preparation strategies, bias analysis, and analytical approach. Novel or insightful interpretations of the data, as well as thoughtful use of basic machine learning to justify data preparation decisions, will be credited.	25%
Total	100%

Some possible online sources for datasets

[1] Kaggle - huge repository of community published data & code - https://www.kaggle.com/datasets

[2] Data.world open datasets - https://data.world/datasets/open-data

[3] Singapore Statistics - https://www.singstat.gov.sg/

[4] Singapore Government Published Dataset - https://data.gov.sg/

[5] Singapore Geo Data dataset - https://data.gov.sg/dataset/national-map-line

[6] US COVID-19 Datasets - https://data.cdc.gov/browse?limitTo=datasets

[7] COVID-19 data - https://github.com/owid/covid-19-data/tree/master/public/data

[8] Our World in Data - Coronavirus Source Data - https://ourworldindata.org/coronavirus-source-data

[9] UCI Machine Learning Repository - https://archive.ics.uci.edu/

[10] Asian Development Bank (ADB) dataset - https://data.adb.org/search/content/type/dataset

[11] DataHub.io Stock Market Data - https://datahub.io/collections/stock-market-data

[12] Google's Dataset Search - https://datasetsearch.research.google.com/

Note: Some sites will require you to sign up as a member to gain access.